

Climate-Controlled Guitar Cabinet

A sealed, humidity-regulated display vault for 8 instruments over an open, vented amp zone — built from scratch in Baltic birch with a cherry frame and a tempered-glass door.

40 × 23 × 85"

EXTERNAL (216 CM · FITS < 220)

45–50%

TARGET VAULT RH

8 + 2

INSTRUMENTS HUNG + COMBO AMPS

~\$2,100

+ \$384 GLASS · EST. MATERIALS



Left: cutaway showing the sealed vault (8 hung instruments, climate gear on the divider) over the open amp zone (two Orange combos + pedalboard). Right: finished cabinet, guitars behind the tempered-glass vault door.

1 · Overview & Goals

Store and display a collection of 8 instruments in a stable, controlled micro-climate while keeping two combo amps and a pedalboard close at hand — in a single floor-standing cabinet that fits a normal room.

Problem

Instruments — especially acoustics with nitrocellulose finishes — are damaged by low humidity (cracking, fret sprout), high humidity (swelling, glue failure, finish hazing), and rapid swings. A normal cabinet does nothing to regulate this. The amps generate heat and must *not* share sealed air with the instruments.

Solution

A two-zone cabinet: a **sealed, cedar-lined upper vault** holding all 8 instruments at 45–50% RH, sitting above an **open, vented lower amp zone** that lets amp heat escape to the room. A single smart controller manages humidity and air circulation and raises alarms; temperature is left to the room (which already sits in range) and watched, not actively driven.

Goals & non-goals

Goals

- Hold vault at **45–50% RH**, $\pm 5\%$, year-round.
- Damp temperature swings; keep within a safe band.
- Display all 8 instruments, visible behind glass.
- Instrument-safe materials and air (low off-gassing).
- Remote visibility + alerts when out of band.
- Fit under an 86.3" / 220 cm ceiling; be tip-safe.

Non-goals

- **Active cooling** — impractical in a sealed wooden box; relies on room ambient.
- Climate control of the **amp zone** — deliberately open/vented.
- Storing the 4 electrics' cases inside — hard cases live elsewhere.
- A furniture-grade finish spec — covered separately.

2 · Recommended (“Ultimate”) Setup

The configuration below is the recommendation after weighing cost, reliability, instrument safety, and control quality.

SUBSYSTEM	DECISION	WHY THIS OVER THE ALTERNATIVE
Shell	From-scratch Baltic-birch carcass ($\frac{3}{4}$ " sides/shelves, $\frac{1}{2}$ " back), 85" tall	Plywood is more dimensionally stable than solid wood and far better than a PAX/particleboard frame; 85" clears the ceiling with margin.
Frame & door	Solid cherry face + door frame; clear tempered, pencil-polished glass (~37×52)	Tempered breaks safe and resists the seamed-edge weakness; pencil-polished edge has no defects. Clear over UV since it sits out of direct sun.
Seal	EPDM gasket + 3 draw latches + piano hinge; airtight caulked seams; cedar lining	A compression seal + cedar buffering is what makes a <i>vault</i> ; magnetic catches can't hold an air seal.
Humidity	Cool-mist humidifier + mini dehumidifier , both on controller outlets	Cool-mist keeps humidity and temperature decoupled (warm-mist would add heat). Two-way control covers dry winters and humid summers.
Control	AC Infinity Controller 79 (outlet model) + independent remote monitor	One unit runs fans (variable speed) + switches humidifier/dehumidifier + logs + alarms. The 69 Pro has no AC outlets; two Inkbirds can't vary fan speed or log.
Temperature	No active control — alarm only	Room sits at 15–25 °C, already in range; the box's thermal mass damps swings. Heating/cooling adds complexity and condensation risk for no benefit.
Air	Two 120 mm variable fans (upper + lower vault)	Even RH/temp, no stratification; variable speed is gentle and quiet vs. on/off.

Why this is the right call (pros)

- Controls the variable that actually matters — **humidity** — accurately, with a single-brain deadband (no fighting).
- **Remote alarms + logging**: you're told before damage happens.
- **Instrument-safe** throughout: Baltic birch, Spanish cedar, low-VOC finish, neutral-cure silicone, distilled-water mist.
- **Display-grade** and **safe**: tempered glass, anchored anti-tip, GFCI power.
- Consolidated wiring; fewer boxes than a multi-Inkbird rig.

Trade-offs (cons) & mitigations

- **DIY effort/skill**, esp. the airtight door → patient gasket fit + leak check.
- **Up-front cost** (~\$2,100 + glass) → most is the shell + glass, both one-time.
- **App/firmware dependency** for logging → independent SensorPush-type monitor as a second, separate alarm; cedar buffers passively if power drops.
- **No active cooling** → acceptable indoors; high-temp alarm + room AC cover the hot edge.

Critical requirement: the door **seal** is the make-or-break. If it leaks, the controllers chase the room all day and never hold setpoint. Budget time for gasket fit, latch tension, and an empty leak/burn-in test before any instrument goes in.

3 · Specifications

Dimensions & zones

External 40 W × 23 D × 85 H in (1016 × 584 × 2159 mm)

Internal 38.5 W × 22 D in

Amp zone (open/vented) 38.5 × 22 × 24 in

Divider top 25.5" from floor

Ceiling clearance fits under 86.3" / 220 cm (~1.3" spare)

Carcass ¾" Baltic birch · ½" back

Vault (sealed) 38.5 × 22 × ~58.75 in

Cable pass-through Ø2" grommet in divider, 2.5" from rear

Environment targets

Vault RH setpoint 45–50% (±5% control band)

Humidify trigger RH below band → humidifier outlet

Deadband setpoints separated so devices never fight

Temperature 15–25 °C ambient — in range, alarm-only

Dehumidify trigger RH above band → dehumidifier outlet

Alarms RH or temp out-of-band → app + independent monitor

Contents (8 instruments + amps)

Hung edge-out in the vault: Egyptian Oud, Taylor GS Mini Bass, slim Martin acoustic, Fender Stratocaster, Schecter C-1 Platinum, Music Man Bass, 1980 Rickenbacker 4003, Martin Backpacker. Amp zone: Orange Rocker 15 (15 W tube) + Orange Crush Bass 50 (50 W solid-state) + pedalboard.

Verify before cutting wood: all instrument dimensions are estimates — measure your own (especially headstock widths and the longest instrument) to confirm the hanging layout and vault clearance.

4 · System Architecture



Labeled system map (from the interactive 3D model). Sealed vault on top; open amp zone below; all climate & electrical gear lives in the vented amp zone, outside the sealed air, with sensing probes passing up through a single sealed grommet.

Operating principle

- **Two thermally/air-separated zones.** The $\frac{3}{4}$ " divider is caulked airtight — it is the vault floor. Everything below it is open to the room.
- **Gear stays out of the sealed air.** Controller, humidifier, dehumidifier, power strip all sit in the amp zone; only thin probes and mist/air routes cross the divider through the sealed grommet (with a drip loop).
- **Closed-loop humidity.** The controller reads vault RH and switches the humidifier or dehumidifier outlet to hold the band, while the two fans keep the vault air mixed.
- **Temperature is observed, not driven.** A high/low alarm is the only temperature action.
- **Independent watchdog.** A separate battery monitor logs RH/temp and alarms even if the main controller fails.

5 · Bill of Materials

Bill of Materials — visual parts list

Every component of the climate cabinet. Icons are schematic. Prices are typical USD estimates.

 Baltic birch ply Carcass ¾" + back ½" Cut to size by dealer part of ~\$885	 Cherry frames 1×2 face + 1×3 door ~4 boards (8') part of ~\$885	 Tempered glass ~37×52, clear Pencil-polished edges ~\$384	 Piano hinge Continuous, 48–60" Stiff door support ~\$25
 Draw latches ×3 Compress door gasket Not magnetic catches ~\$20	 Door pull Bar/knob handle Your finish choice ~\$12	 Guitar hangers ×8 Wood wall-mount (String Swing-type) ~\$145	 Spanish cedar ¼" vault lining ~35–40 sq ft ~\$150
 AC Infinity 79 Outlet controller Fans+RH+app+alarms ~\$110	 Circulation fans ×2 120 mm, variable Run by the 79 ~\$70	 Cool-mist humidifier Distilled water On a 79 outlet ~\$40	 Mini dehumidifier Peltier / desiccant On a 79 outlet ~\$45
 Remote monitor SensorPush-type Independent RH alarm ~\$50	 Power & cabling GFCI strip, grommet, raceway, drip loops ~\$65	 Sealing kit Caulk, neutral silicone, EPDM, glazing tape ~\$70	 Anti-tip brackets Anchor to studs Safety-critical ~\$12

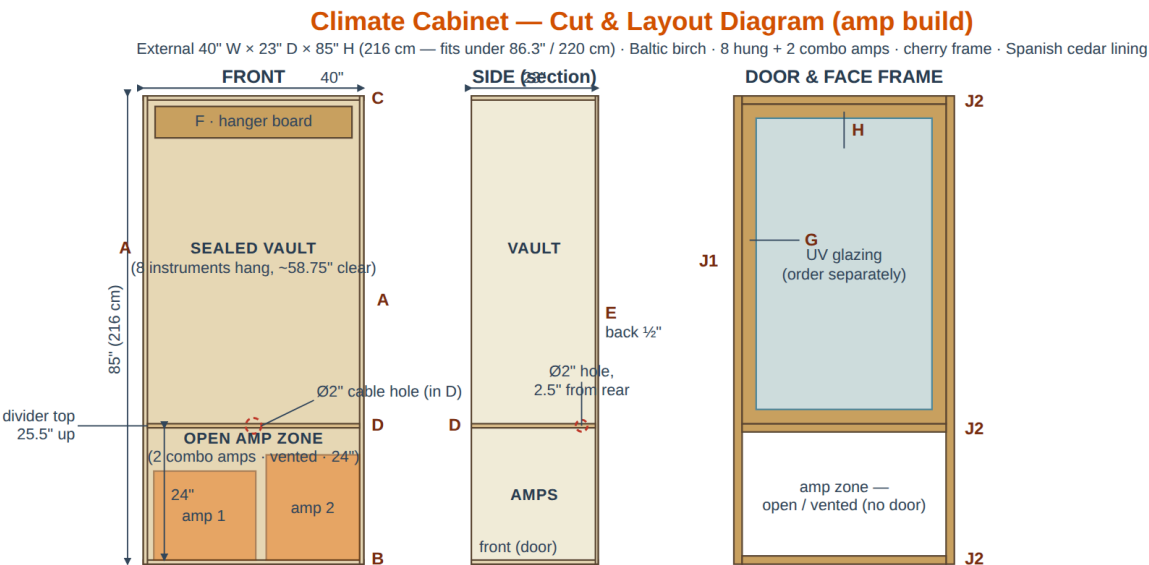
Visual parts list. Detailed quantities, specs, and estimated costs below.

ITEM	SPEC / NOTES	QTY	EST. COST
Shell & frame — cut to size by a hardwood dealer			
¾" Baltic birch plywood	Sides A×2 (22.5×85), top/bottom/divider B/C/D, hanger F	2 sheets	incl.
½" Baltic birch plywood	Back panel E (40×85)	1 sheet	incl.
Cherry 1×2 + 1×3	Face frame (1×2) + vault door frame (1×3), 8' boards	~4	incl.
Ø2" grommet hole	Bored in divider, 2.5" from rear	1	incl.
<i>Dealer quote (cut Baltic birch + cherry)</i>			~\$885
Door, glazing & seal			
Tempered glass, clear	~37×52, pencil-polished edges, cut to size	1	\$384
Foam glazing tape	Beds glass in the rabbet	1 roll	\$8

Piano hinge	Continuous, 48–60" (or 3 cabinet hinges)	1	\$25
Draw / compression latches	Clamp door onto gasket — not magnetic	3	\$20
Door pull	Bar or knob	1	\$12
EPDM / silicone weatherstrip	Door gasket — the key seal	1 roll	\$15
Interior & hanging			
Spanish cedar lining	¼", vault walls + back + floor (~35–40 sq ft)	1	\$150
Brad nails / construction adhesive	Attach cedar	1	\$15
Guitar hangers	Wood wall-mount (String Swing-type)	8	\$145
Climate & control			
AC Infinity Controller 79	Outlet controller: fans + RH outlets + app + alarms	1	\$110
Circulation fans (UIS)	120 mm variable, upper + lower vault	2	\$70
Cool-mist humidifier	Ultrasonic/evaporative, run on distilled water	1	\$40
Mini dehumidifier	Peltier or renewable desiccant	1	\$45
Remote monitor	SensorPush-type independent RH/temp alarm	1	\$50
Optional: Inkbird ITC-308	Independent temperature alarm / future heat switch	1	+\$35
Power, sealing & safety			
GFCI power strip	Resettable breaker, surge	1	\$25
Cable grommet (2") + duct-seal	Sealed pass-through; re-workable putty around cables	1	\$13
Cable management	Raceway, ties, drip loops	1	\$15
Acrylic-latex caulk	Airtight vault seams (low-odor)	3	\$15
Neutral-cure silicone	Beds glass — not acetoxxy (corrodes strings)	1	\$8
Low-VOC water-based poly + brush	Seal interior faces not cedar-lined	1 qt	\$30
Anti-tip wall brackets	Anchor to studs — safety-critical	1 set	\$12
Wood glue + pocket screws	Carcass assembly (if self-building)	1	\$25
Estimated materials total (excl. optional Inkbird & tools)			~\$2,100

Costs are typical USD estimates and exclude shipping, tools, and the amps/instruments (already owned). The dealer line consolidates all cut plywood + cherry into the single ~\$885 quote.

6 · Cut List & Layout



PARTS LIST (dimensions in inches; W × H for uprights, W × D for shelves)

Baltic birch plywood — ¾" (carcass) & ½" (back)

- A · Side panel ×2 — 22.5 D × 85 H — ¾" Baltic birch
- B · Bottom ×1 — 38.5 W × 22.5 D — ¾" Baltic birch
- C · Top ×1 — 38.5 W × 22.5 D — ¾" Baltic birch
- D · Divider / middle shelf ×1 — 38.5 W × 22.5 D — ¾" Baltic birch
- E · Back ×1 — 40 W × 85 H — ½" Baltic birch
- F · Hanger board ×1 — 36 W × 6 H — ¾" Baltic birch

Pre-cut hole

Ø2" in Divider D — centered, 2.5" from rear edge

Solid hardwood — CHERRY (door + face frame)

- G · Door stile ×2 — 2.5 W × 57 L (trim to fit)
- H · Door rail ×2 — 2.5 W × 35 L (trim to fit)
- J1 · Face-frame stile ×2 — 1.5 W × 85 L
- J2 · Face-frame rail ×3 — 1.5 W × 37 L

Interior lining — Spanish cedar (~¾")

Line vault walls + back + floor · fit/trim on site · leave bare

Notes

Glazing (UV acrylic/glass ~37 × 52) ordered separately — not wood.
Optional foil/foam thermal break under D (amps below vault).

Cut & layout diagram — front elevation, side section, and door/face-frame, with the full parts legend. Hand this to the hardwood dealer.

#	PART	QTY	DIMENSIONS	MATERIAL
A	Side panel	2	22.5 D × 85 H	¾" Baltic birch
B	Bottom	1	38.5 W × 22.5 D	¾" Baltic birch
C	Top	1	38.5 W × 22.5 D	¾" Baltic birch
D	Divider shelf (vault floor)	1	38.5 W × 22.5 D · Ø2" hole	¾" Baltic birch
E	Back	1	40 W × 85 H	½" Baltic birch
F	Hanger board	1	36 W × 6 H	¾" Baltic birch
G	Door stiles	2	2.5 × ~57	Cherry 1×3
H	Door rails	2	2.5 × ~35	Cherry 1×3
I	Door glazing	1	~37 × 52	Tempered glass

J	Face frame	2+3	1.5 × 85 (stiles) · 1.5 × 37 (rails)	Cherry 1×2
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Plywood nests onto two ¾" sheets + one ½" sheet; cherry from ~4 8' boards. The two sides fit on one ¾" sheet (45" of 48" width); top/bottom/divider/hanger on the second.

7 · Build Sequence

1. **Confirm & order.** Measure real plywood thickness; hand the cut list to the dealer (incl. the Ø2" divider hole). Order cherry. Hold the glass order until the door is built.
2. **Pre-finish vault interior.** Sand and apply 2 coats low-VOC water-based poly to inner faces you won't cedar-line; let cure fully (it off-gasses).
3. **Assemble carcass.** Bottom, then divider at the 25.5" line, then top, housed between the full-height sides; glue + pocket screws; check square; glue + screw the back.
4. **Sealed pass-through.** Fit the grommet in the divider hole; bead caulk around it; this is the only route into the vault.
5. **Caulk the vault airtight.** Every interior seam above the divider. Leave the amp zone open — never seal it.
6. **Line the vault with cedar.** Walls + back + floor; brad nails / adhesive; leave the door opening clear.
7. **Hanger board + hangers.** Mount F to the back; fit the 8 hangers to the measured layout (edge-out, height-staggered).
8. **Build & hang the door.** Cherry frame with a rabbet; set glass on glazing tape + a neutral-cure silicone bead; hang on the piano hinge; add the EPDM gasket and 3 draw latches so closing *compresses* the seal.
9. **Install gear.** Controller, humidifier, dehumidifier, GFCI strip in the amp zone; fans + sensor in the vault; route probes through the grommet with drip loops.
10. **Anchor anti-tip** to wall studs.
11. **Burn-in & calibrate.** Run empty 2–3 days; salt-calibrate the sensor; set the RH band + deadband; leak-check the door; confirm it holds 45–50% RH before instruments go in.

8 · Climate & Control Logic

CONDITION	ACTION	DEVICE
Vault RH < 45% (low edge)	Energize humidifier outlet; run fans	Humidifier (cool-mist) + fans
Vault RH > 50% (high edge)	Energize dehumidifier outlet; run fans	Mini dehumidifier + fans
RH inside band	Both moisture devices off; fans cycle for mixing	Fans only
Temp < 15 °C or > 25 °C	Alarm (app + monitor). No active heat/cool.	Controller + remote monitor
RH out of band > set time	Push alarm; log event	Controller + remote monitor
Power loss	Cedar buffers passively; monitor (battery) keeps logging/alarming	Cedar + independent monitor

Why cool-mist: warm-mist humidifiers are essentially small heaters (~200–400 W) and would couple humidity to temperature in this small box. Cool-mist keeps the two independent — humidity from the humidifier, temperature left to the room. Use **distilled water** to avoid mineral dust on finishes.

Deadband rule: separate the humidify and dehumidify setpoints (e.g. on at 44% / off at 47% for humidify; on at 51% / off at 48% for dehumidify) so the two devices can never run against each other.

9 · Requirements & Acceptance Criteria

ID	REQUIREMENT	ACCEPTANCE TEST
R1	Vault holds 45–50% RH	Empty 48 h burn-in stays in band $\pm 5\%$ with no manual intervention.
R2	Air seal is effective	With humidifier off, vault RH decays toward room only slowly (hours, not minutes); no obvious leak path at the door perimeter.
R3	Temperature stays in band	Logged vault temp remains 15–25 °C across a normal day; high/low alarm fires when forced out.
R4	Devices never fight	Logs show humidifier and dehumidifier never energized within the same deadband cycle.
R5	Amp heat is exhausted	With both amps on, amp-zone heat vents to the room; vault temp unaffected (sealed divider).
R6	Independent alarm path	Remote monitor alerts on out-of-band RH/temp with the main controller unplugged.
R7	Structural safety	Cabinet anchored to studs; does not tip when the door is opened fully and loaded.
R8	Material safety	All finishes/sealants fully cured before loading; no acetoxysilicone; distilled water only.
R9	Fits the room	Stands upright under the 86.3" / 220 cm ceiling and walks into final position.

10 · Decisions Log & Open Items

Locked decisions

DECISION	RESOLUTION
Build vs. retrofit	Build from scratch (Baltic birch) — better seal & stability than a PAX/retrofit.
Height	85" (216 cm) — extra room into the vault while clearing the ceiling.
Amps inside?	Yes — open, vented 24" amp zone below the vault.
Glazing	Clear tempered, pencil-polished edges (all edgework free; chose the strongest/safest).
Humidifier type	Cool-mist (keeps temperature independent).
Controller	AC Infinity Controller 79 (outlet model) over 69 Pro / dual Inkbird.
Active temperature control	Dropped — 15–25 °C is in range; alarm only.
Thermal break under divider	Optional — amps run only briefly.

Open items

- **Confirm ceiling height** is a true 86.3" after molding/baseboard, leaving room to tilt the cabinet upright.
- **Measure instruments** (headstock widths, longest body) to finalize the 8-up hanging layout before cutting.
- **Combo humidify/dehumidify unit?** If a single self-regulating combo is chosen, it manages RH itself and the 79 reduces to fans + monitoring — confirm it does both automatically with an accurate humidistat.
- **Order glass last** — only after the cherry door is built and the opening measured (tempered can't be trimmed).
- **Confirm Controller 79 outlet count** (≥ 2 switchable) and that it has a humidity trigger.